# Anqi Tan

Ph.D. Student in Bioengineering · University of Illinois Urbana-Champaign

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## EDUCATION

**Ph.D. in Bioengineering, University of Illinois Urbana-Champaign** Aug 2022 – Present
*Advisor: Prof. Brian T. Cunningham*

**M.S. in Bioengineering, University of Illinois Urbana-Champaign** Aug 2022 – Present

**B.S. in Mechanical Engineering, University of Illinois Urbana-Champaign** Aug 2018 – Jun 2022

**B.S. in Mechanical Engineering, Zhejiang University (ZJU-UIUC Institute)** Aug 2018 – Jun 2022

## RESEARCH INTERESTS

Biosensing Devices · Point-of-Care Diagnostics · Portable Optical Readers & Instrumentation · Hardware–Software Integration for Biomedical Measurement · Plasmon-Enhanced Lateral Flow Tests · Real-Time Hormone Monitoring

## RESEARCH EXPERIENCE

**Graduate Research Assistant, Nanosensors Group, UIUC** Sep 2022 – Present

- Leading development of the portable fluorescence reader for a $4.87M DoD (PRMRP) project on point-of-care phenotyping of sepsis and ARDS, in collaboration with Stanford, WashU, and UCSF.
- Leading development of a pocket-size scanner for plasmonic-fluor-tagged lateral flow test strips (first-author manuscript in preparation).
- Leading the Live Hormone project: a competitive LFA platform for real-time testosterone, cortisol, and progesterone monitoring.
- Contributing to plasmon-enhanced LFAs for infectious disease detection and to reviews in Chemical Reviews and Lab on a Chip.

**Research Assistant Intern, Institute of Brain Science, Zhejiang Qiangnao Technology Co., Ltd. (BrainCo), Hangzhou** May 2022 – Aug 2022

- Completed research tasks for two projects: sleep assisted by acoustic and electrical stimulation, and fatigue monitoring with multimodal EEG.

**Undergraduate Research Leader, ZJU-UIUC Institute, Zhejiang University** 2019 – 2021

- Led a national-level research training project on a re-localizable RCM mechanism for minimally invasive surgery (first-author paper, IEEE ICARM 2021).
- Led a team implementing camera tracking for a robot-manipulated endoscope; contributed to a 3D-printed concrete dragon boat (Structures20/ACEM20).

**Senior Design (ECE 445, Team #29): Mimicry Stage Lighting Control System** Feb 2022 – May 2022

- Mechanical design lead: two-axis light manipulator (MATLAB forward/inverse kinematics; Euler–Lagrange and Newton–Euler torque analysis) and a 3D-printed, laser-cut structure designed for assembly and portability (9.3 kg, 4.5 h battery life).
- A handheld ESP32 + MPU9250 joystick drives the light in real time over WiFi; a music-analysis module generates LED color sequences.

## RESEARCH FUNDING PARTICIPATION

**DoD Peer Reviewed Medical Research Program, Award HT9425-25-1-0825 ($4.87M)** 2025 – Present

- Point-of-care molecular phenotyping of sepsis and ARDS (PI: Pratik Sinha; with S. Singamaneni, C. Calfee, and B. T. Cunningham). Role: lead developer of the portable fluorescence reader.

## PUBLICATIONS

***Journal Articles and Reviews***

1. Seemesh Bhaskar, Saurabh Umrao, Han Keun Lee, Joseph Tibbs, Amanda Bacon, Skye Shepherd, Takhmina Ayupova, Fatma Uysal Ciloglu, Leyang Liu, **Anqi Tan**, Wang-Chien Chen, My Thi Tra Nguyen, Maria Grace Scannell, Ugur Aygun, Ugur Parlatan, Catherine Zhang, Manish Kohli, Guy R. Adami, Wali Badar, Ron C. Gaba, Aaron Mansfield, Joel Schwartz, Xing Wang, Utkan Demirci, and Brian T. Cunningham. "Point of Care Molecular Cancer Diagnostics." *Lab on a Chip, 26(10), 2948–3001. 2026.*
2. Yuxiong Liu, Qisheng Jiang, **Anqi Tan**, Gayathri Kondepati, Prashant Gupta, Yixuan Wang, Yajur Chandar, Tracey H. Bach, Stacey L. House, Peter A. Rice, Bijal A. Parikh, Brian T. Cunningham, Sanjay Ram, Ige A. George, Srikanth Singamaneni, and Sumanth Gandra. "Ultrasensitive Point-of-Care Antigen Test for Detection of Neisseria gonorrhoeae Using Plasmon-Enhanced Lateral Flow Assays." *ACS Nano Medicine, 1(4), 1036–1047. 2026.*
3. Seemesh Bhaskar, Weijing Wang, Hankeun Lee, Leyang Liu, Saurabh Umrao, Weinan Liu, Amanda Bacon, Joseph Tibbs, Kodchakorn Khemtonglang, **Anqi Tan**, Takhmina Ayupova, Wang-Chien Chen, Xing Wang, and Brian T. Cunningham. "Photonic Crystal Grating Resonance and Interfaces for Health Diagnostic Technologies." *Chemical Reviews, 125(14), 6435–6540. 2025.*
4. Amanda Bacon, Weijing Wang, Hankeun Lee, Saurabh Umrao, Prima Dewi Sinawang, Demir Akin, Kodchakorn Khemtonglang, **Anqi Tan**, Sabina Hirshfield, Utkan Demirci, Xing Wang, and Brian T. Cunningham. "Review of HIV Self Testing Technologies and Promising Approaches for the Next Generation." *Biosensors, 13(2), 298. 2023.*

***Book Chapters***

1. Seemesh Bhaskar, Yanyu Xiong, Skye Shepherd, Joseph Tibbs, Amanda K. Bacon, Weinan Liu, Lucas D. Akin, Takhmina Ayupova, Weijing Wang, Hankeun Lee, Leyang Liu, **Anqi Tan**, Kodchakorn Khemtonglang, Xiaojing Wang, and Brian T. Cunningham. "Nano-Engineering at Functional Photonic Crystal Interfaces." *In Nano-Engineering at Functional Interfaces for Multi-Disciplinary Applications, Elsevier. 2025.*

***Conference Papers and Abstracts***

1. **Anqi Tan**, Kai-Ning Wang, Tasfia Ashrafee, Ami Patel, Yuxiong Liu, Saurabh Umrao, Srikanth Singamaneni, Jenna L. Riis, Jacinda K. Dariotis, and Brian T. Cunningham. "Portable Plasmonic Fluor-Enhanced Lateral Flow Assay for Quantitative Salivary Cortisol Measurement in Community Settings." *APHA 2026 Annual Meeting and Expo, San Antonio, TX. Accepted for oral presentation, Nov 3, 2026.*
2. **Anqi Tan**, Shenwei Chen, Hanyu Gan, and Liangjing Yang. "Design Optimization of a Mechanically Constrained Re-Localizable Remote-Center-of-Motion." *2021 6th IEEE International Conference on Advanced Robotics and Mechatronics (ICARM). 2021.*
3. Y.Q. Feng, K.H. Zhang, **A.Q. Tan**, K. Ma, and Y. Xiao. "Utilization of 3D Printing for Concrete Dragon Boat Model." *The 2020 Structures Congress (Structures20), in the 2020 World Congress on Advances in Civil, Environmental, & Materials Research (ACEM20), Seoul, Korea, Aug 25–28, 2020.*

***Manuscripts Under Review / In Preparation***

1. **Anqi Tan**, Han Keun Lee, Yuxiong Liu, Liang-ching Chen, Leyang Liu, Siyi Li, Xiaoyang Chu, Saurabh Umrao, Xing Wang, Srikanth Singamaneni, and Brian T. Cunningham. "Pocket Size Lateral Flow Assay Scanner for Plasmonic Fluor Tagged Paper Test Strip Assays." *Manuscript in preparation.*

## CONFERENCE PRESENTATIONS

1. Oral presentation (accepted): "Portable plasmonic fluor-enhanced lateral flow assay for quantitative salivary cortisol measurement in community settings." APHA 2026 Annual Meeting and Expo, San Antonio, TX, Nov 3, 2026. (Nov 2026)
2. Poster: "A portable plasmonic fluor-enhanced lateral flow platform for quantitative salivary cortisol measurement." Lab-on-a-Chip and Microfluidics World Congress 2026 (SelectBIO), Laguna Hills, CA. (Oct 2026)

3. Poster: "Portable plasmonic fluor-enhanced lateral flow assay for quantitative salivary cortisol measurement in community settings." Health Care Engineering Systems Symposium 2026 (Jump ARCHES), Peoria, IL. (Oct 2026)
4. Poster (co-author): "Assessing the acceptability and feasibility of real-time salivary hormone feedback and implications for building trust in biomedical research: A mixed-methods study." Health Care Engineering Systems Symposium 2026 (Jump ARCHES), Peoria, IL. (Oct 2026)
5. Poster: "Portable plasmonic fluor-enhanced lateral flow assay for quantitative salivary cortisol measurement in community settings." 2026 Personalized Nutrition Innovation Day, UIUC. (Sep 2026)
6. Poster (co-author): "Assessing the acceptability and feasibility of real-time salivary hormone feedback and implications for building trust in biomedical research: A mixed-methods study." 2026 Personalized Nutrition Innovation Day, UIUC. (Sep 2026)
7. Poster: "Validating a lateral flow assay (LFA) scanning device for real-time salivary cortisol assaying." 2025 Personalized Nutrition Innovation Day, UIUC. (Oct 2025)

## HONORS AND AWARDS

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| First Place ($3,000), Ashby Prize in Computational Science Hackathon, NCSA, UIUC | 2025 |
| Honorable Mention, Ashby Prize in Computational Science Hackathon, NCSA, UIUC | 2024 |
| National Silver Medal, China "Internet+" Innovation and Entrepreneurship Competition | 2021 |
| First Prize, RoboMaster Mecha Master (College Level) | 2019 |
| Outstanding Performance and Innovation Award, 1st International Concrete Dragon Boat Competition (ICDBC), American Concrete Institute & Zhejiang University | 2019 |
| Third-Class Scholarship, Zhejiang University | 2019 |
| Public Service Pacesetter, Zhejiang University | 2019 |
| Model of Social Work, Zhejiang University (twice) | 2019, 2020 |
| Second Place, Social Research Competition, as team leader (three times) | 2019 – 2020 |
| Second Prize, Zhejiang Province Biology Olympiad | 2017 |

## TEACHING

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| Teaching Assistant, BIOE/ECE 415 Biomedical Instrumentation Lab, UIUC | Fall 2025 |
| Teaching Assistant, ME 170 Computer-Aided Design, UIUC | Spring 2020, Spring 2021, Spring 2022 |

## LEADERSHIP AND SERVICE

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| President, Residential College Student Committee, Zhejiang University, Haining Campus | 2019 – 2022 |
| Volunteer Teacher, Rural Education Program, Guizhou Province | Aug 2019 |

## SKILLS

**Biosensing & Wet Lab:** Lateral Flow Assays, Plasmonic-Fluor Labels, ELISA, 3D Cell Culture, Fluorescence Staining

**Mechanical Design & CAD:** SolidWorks, Creo, Inventor, Fusion 360, AutoCAD

**Fabrication & Simulation:** 3D Printing, Melt Electrowriting, COMSOL, FEA

**Programming & Hardware:** Python, MATLAB, LabVIEW, Arduino

**Languages:** English (Fluent), Chinese (Native)